Chlorine Decay in Municipal Storage Tanks

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FICTIONAL TEST MATERIAL: this paper does not exist; it was written for the citation-audit fixture. No author, utility, venue, or measurement described here is real.

Abstract

Treated water loses its free chlorine while it sits in storage tanks, and a tank that holds water too long can deliver water below the residual a utility must maintain. We sampled eight storage tanks of a fictional municipal system weekly for a year. Free chlorine decayed at 0.08 to 0.15 mg/L per day, and the decay was faster in the warm months than in the cool months. We relate the decay rate to water age and give a simple rule for the maximum residence time that keeps the residual above its target.

1 Introduction

Utilities add chlorine at the treatment works so that a free-chlorine residual remains at every tap. Between the works and the customer, the water often sits in service reservoirs and elevated tanks. During that time, chlorine reacts with organic matter in the water and with the tank walls, and the residual falls.

The rate of that fall decides how long water may stay in a tank. This paper measures it in eight tanks of the fictional Fixture Lake system over a full year.

2 Method

2.1 Tanks and sampling

The eight tanks range from 500 to 12,000 cubic metres, and include ground-level reservoirs and elevated towers. Each week, a sampler took water from the tank inlet and from the tank outlet and measured free chlorine on site with a colorimetric kit. The tank-water temperature was recorded at each visit.

2.2 Decay rate

For each visit, the decay rate is the inlet-to-outlet drop in free chlorine divided by the mean residence time of the water in the tank, which we estimated from the tank volume and the logged inflow. We report rates in milligrams per litre per day.

3 Results

3.1 Decay rates

Free chlorine in the storage tanks decayed at 0.08 to 0.15 mg/L per day. The lowest rates came from the large ground-level reservoirs, and the highest from the small elevated towers, which have more wall area per unit volume.

3.2 Seasonal pattern

Decay was faster in the warm months. In every tank, the rate measured between June and September exceeded the rate measured between December and March. The warm-month rates clustered near the top of the range, at 0.13 to 0.15 mg/L per day, and the cool-month rates near the bottom, at 0.08 to 0.10 mg/L per day.

3.3 Residence-time rule

With an inlet residual of 0.8 mg/L and a target outlet residual of 0.2 mg/L, the measured rates allow a residence time of about four days in the warm months and about six days in the cool months. Operators can meet the target by drawing tanks down further in summer.

4 Limitations

This study measures chlorine decay in storage tanks only.

This study does not examine pipe bursts or any other mechanical failure of the network: no main breaks were recorded and none were analysed. This study does not examine temperature effects on pipes or on the surrounding soil; the only temperature we measured was that of the water inside the tanks. This study does not examine cold weather: the coldest tank-water temperature we recorded was 9 °C, and no sampling took place under freezing conditions. Decay in the distribution mains downstream of the tanks was not measured.